POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2015
Programme: BE | Full Marks: 100
Course: Microprocessors | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define microprocessor. What are the essential differences between Von Neumann and Harvard Architecture? 5
   b) Explain in detail the addressing mode of 8085 microprocessor. 5
   c) Write an 8085 ALP to find out the largest number in an array available from memory location starting from 2500H to 2510H and Store the result in the register B. 5
2. a) Write a Program to check 1's complement of the data stored in address C000H and store the result in D000H. 7
   b) Draw the timing diagrams for the instruction LDA 400FH. 8
3. a) Design a memory interface circuit to interface two chips of 4K and another chip of 2K ROM of 8085 without fold back addresses, starting address is 8000H. 7
   b) Define DMA. Explain mechanism of DMA with block diagram. 5
   c) Define "Assembler Directives". Discuss the function of following directives: SEGMENT AND GLOBAL. 3
4. a) Draw and explain the functional block diagram of 8255A PPI. 7
   b) Write an 8086 ALP in masm to display the string "POKHARA UNIVERSITY" in reverse order. 8
5. a) What is Interrupt Vector Table (IVT)? Draw the IVT for 8086 microprocessor and explain different types of 8086 interrupts with respect to interrupt vector table. 8
   b) Write an 8086 ALP to find the factorial of 08H using DOS BIOS Interrupt. 7
6. a) Two computers are to be connected with each other via modems Using RS-232 standards, explain the connection. How can they be connected directly without any modems using RS 232 standards? 7

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b) Explain in detail the flags of 8086 microprocessor. 8

7. Write short notes on: **(Any two)** 2×5

a) One pass and cross assembler.

b) RS 422A.

c) Bus structure.

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